

Lab 06 Tasks

Lab Tasks (Lab Timing)

Computer Screen Detection in a Computer Lab

You are working on a computer vision project for monitoring computer lab usage and ensuring the availability of computer screens. Your goal is to implement screen detection using the Hough Line Transformation. The lab contains rows of computers, and you need to identify the boundaries of computer screens to monitor their status (on or off) and detect any anomalies, such as missing screens.

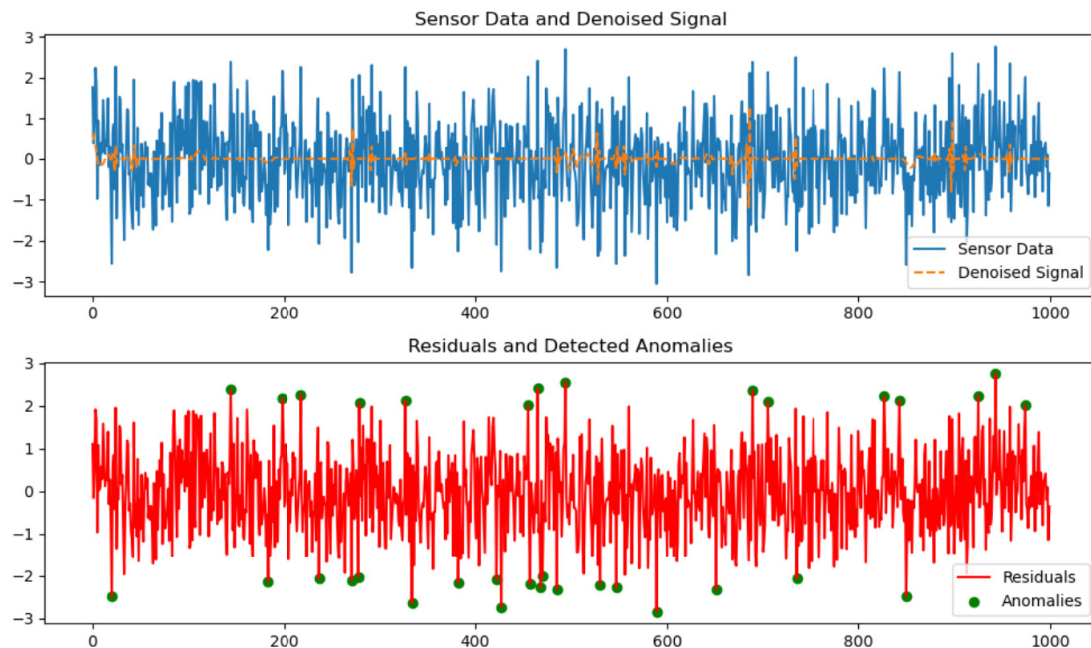
Asset Tracking in a Computer Lab Using SIFT

You are responsible for managing computer assets in a busy computing lab. One of the challenges you face is keeping track of the various computers and their components within the lab. To streamline this process, you want to implement a computer vision solution using the Scale-Invariant Feature Transform (SIFT). Your goal is to automatically recognize and identify individual computer systems and their unique components, such as monitors and keyboards, to maintain an up-to-date inventory

Anomaly Detection in Sensor Data Using Wavelet Transformation

You are tasked with developing a system to monitor sensor data from industrial machines and identify anomalies in real-time. Anomalies may indicate potential machine malfunctions or issues. Your approach is to use the wavelet transformation to analyze the sensor data and detect abnormal patterns. Your goal is to create a simple anomaly detection system using a sample sensor dataset and wavelet analysis.

Output Expected



Lab Tasks

Object Recognition (Using Video)

You are working on an image processing project, and your task is to implement object recognition using the Scale-Invariant Feature Transform (SIFT) algorithm. You have a reference image of the object you want to recognize and a set of test images. Your goal is to identify and locate the object in each test image, even if it appears at different scales, orientations, or under partial occlusion. You will use SIFT features for object recognition and draw bounding boxes around the recognized objects in the test images.

Your Panoramic Image

You are working on a project where you need to create a panoramic image by stitching multiple overlapping images together. The task involves taking a set of images captured from a camera while panning, and then using the SIFT algorithm to find key points and match them between adjacent images. Your goal is to automatically align and stitch the images to create a single panoramic image.

Lane detection using the Hough Line Transformation

You are working on an autonomous vehicle project, and one of the critical tasks is to detect lane markings on the road to ensure safe

and accurate navigation. Your goal is to implement lane detection using the Hough Line Transformation. You have a set of images captured from the vehicle's camera. Your task is to identify and draw the detected lane lines on these images to assist in lane-keeping and control algorithms.

Coins Detection and Counting

You are working on a computer vision project that involves the identification and counting of coins in images. Your task is to implement coin detection using the Hough Circle Transformation. You have a set of images containing coins of various sizes and positions. Your goal is to automatically identify and count the coins in these images.

Smart Security System

You are working on a security system for an organization, and one of the key tasks is to identify unauthorized objects that are placed near sensitive areas. To achieve this, you need to implement boundary detection using computer vision. Your goal is to detect the boundaries of objects that are within a predefined security zone in a real-time video stream. When an unauthorized object enters this zone, an alarm should be triggered.